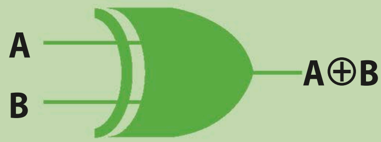
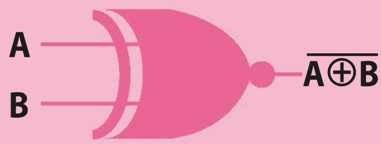
Logic gate	Symbol	Truth table

XOR
An XOR gate's output is 1 only when both its inputs are different. This is sometimes known as Exclusive-OR.



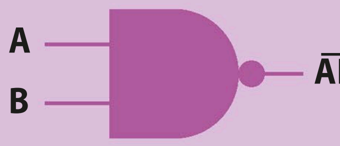
INPUT		OUTPUT
A	B	A XOR B
0	0	0
0	1	1
1	0	1
1	1	0

XNOR
An XNOR gate is equivalent to an XOR gate followed by a NOT gate. Its output is 1 only when both its inputs are the same.



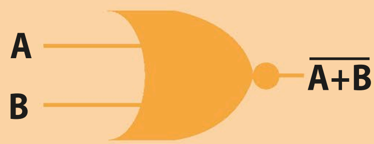
INPUT		OUTPUT
A	B	A XNOR B
0	0	1
0	1	0
1	0	0
1	1	1

NAND
A NAND gate is equivalent to an AND gate followed by a NOT gate. Its output is 1 unless both its inputs are 1.



INPUT		OUTPUT
A	B	A NAND B
0	0	1
0	1	1
1	0	1
1	1	0

NOR
A NOR gate is equivalent to an OR gate followed by a NOT gate. Its output is 1 only when both its inputs are 0.

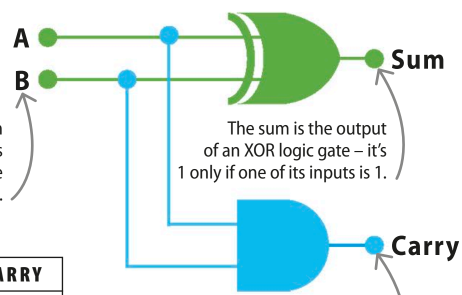


INPUT		OUTPUT
A	B	A NOR B
0	0	1
0	1	0
1	0	0
1	1	0

Combining gates

Computers carry out addition using circuits called binary adders. As computers do everything in 0s and 1s, binary adders must be combined to add numbers of several digits. They do this by having two outputs, sum and carry. The sum is the result of adding the inputs together. If both are 1, the output should be 2, which in binary is 10. The carry output becomes 1 to move the digit representing 2 into the next adder along.

▷ **Half-adders**
Each adder contains two smaller circuits called half-adders made of only two logic gates, an XOR and an AND.



The inputs (A and B) to the sum gate are the same as the inputs to the carry gate as they are connected together.

The sum is the output of an XOR logic gate – it's 1 only if one of its inputs is 1.

The carry output is an AND logic gate. Its output is 1 only if both inputs are 1. It carries a value over to the next adder when adding large numbers.

A	B	SUM	CARRY
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1